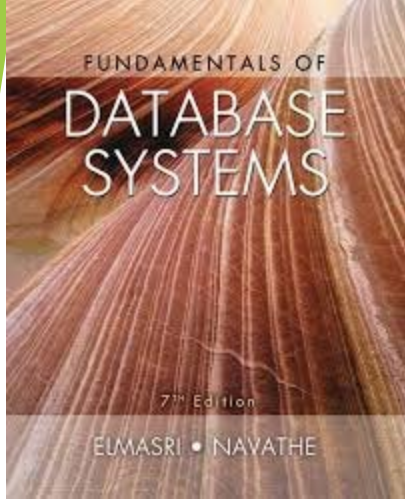




Chapter 1

Databases and Database Users

Fundamentals of Database Systems

7th Edition

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8.6.2026

Lecture 1

What is a Database?

A **database** is a collection of related data.

Data are facts that can be recorded and have meaning.

Examples include names, phone numbers, addresses, student records, and employee information.

Example:

- Contact list stored in a mobile phone
- Address book
- Student records in a school

Importance of Databases

Databases play an important role in many fields where computers are used, such as:

- Business
- E-commerce
- Social Media
- Engineering
- Medicine
- Education
- Law
- Library Science

They help organizations store, manage, and retrieve information efficiently.

Main Characteristics of a Database

1. Represents the Real World

A database represents a part of the real world, called the **Miniworld** or **Universe of Discourse (UoD)**.

Example:

A university database stores information about students, courses, and teachers.

When a student enrolls in a new course, the database must be updated.

2. Logically Organized Data

A database contains data that are logically related and meaningful.

Example:

A student database containing Student ID, Name, and Department is meaningful.

A random collection of unrelated data is **not** considered a database.

3. Designed for a Specific Purpose

A database is created, maintained, and used for a particular purpose by a specific group of users.

Examples:

- Hospital Database → Patient records
- Bank Database → Customer accounts
- Library Database → Book information

Database Updates

A database must reflect changes occurring in the real world.

Examples:

- Customer purchases a product.
- Employee information changes.
- Student registers for a course.

To remain accurate, the database should be updated whenever changes occur.

Database Size and Complexity

Databases can be:

Small Database

- Personal contact list
- Address book
- Few hundred records

Large Database

- Library catalog with thousands of books
- University database with thousands of students

Very Large Database

- Social media platforms
- Online shopping websites

These databases contain millions or billions of records and require powerful systems to manage them.

Examples of Large Databases

Social Media Database

Stores:

- User profiles
- Friend relationships
- Posts and comments
- Privacy settings

Online Shopping Database

Stores:

- Customer information
- Product details
- Orders
- Inventory data

These databases are continuously updated as users interact with the system.

Types of Databases

1. Manual Database

Stored and maintained manually.

Example:

- Library card catalog
- Paper records

2. Computerized Database

Stored and managed using computers.

It can be maintained by:

- ✓ Application programs
- ✓ Database Management Systems (DBMS)

Examples:

- MySQL
- Microsoft Access
- Oracle
- SQL Server

What is a DBMS?

A **Database Management System (DBMS)** is software that allows users to **create, store, manage, and maintain databases**.

It acts as an interface between users/applications and the database.

Definition:

A DBMS is a computerized system that enables users to create and maintain a database.

Examples of DBMS:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- Microsoft Access

Main Functions of a DBMS

A DBMS performs four major functions:

1. Defining a Database

Defining a database means specifying:

- Data types
- Data structures
- Constraints

The DBMS stores this information in a **Database Catalog (Data Dictionary)**.

This descriptive information is called **Metadata**.

Metadata means "**data about data**."

Field Name	Data Type	Size
StudentID	Integer	10
Name	Varchar	50
Age	Integer	3

The above table structure is metadata.

2. Constructing a Database

Constructing a database means storing data on storage devices controlled by the DBMS.

Example

Student Table

StudentID	Name	Major
101	Aung Aung	CS
102	Su Su	IT

The DBMS stores and organizes these records efficiently.

3. Manipulating a Database

Manipulation means performing operations on the database.

Common operations include:

(a) Querying

Retrieving specific information.

Example

```
SELECT * FROM Student;
```

(b) Updating

Modifying existing data.

Example

UPDATE Student

SET Major='Computer Science'

WHERE StudentID=101;

(c) Generating Reports

Producing formatted information for users.

Example

- Student List
- Employee Salary Report
- Monthly Sales Report

4. Sharing a Database

A DBMS allows multiple users and applications to access the database simultaneously.

Example

In a university system:

- Students view results.
- Lecturers enter marks.
- Administrators manage records.

All users can work on the same database at the same time.

Queries and Transactions

1. Query

A query is a request to retrieve information from a database.

Example

SELECT Name

FROM Student

WHERE Major='IT';

Result: **Names of all IT students.**

2. Transaction

A transaction is a logical unit of work that may:

- Read data
- Insert data
- Update data
- Delete data

Eg: Bank Transfer

1.Deduct money from Account A

2.Add money to Account B

Both operations must be completed successfully.

Additional Functions of a DBMS

1. Database Protection

A DBMS protects the database from:

Hardware Failures

- Power failure
- Hard disk crash

Software Failures

- Program errors
- System crashes

The DBMS can recover lost data using backup and recovery mechanisms.

2. Security Protection

The DBMS prevents unauthorized access.

Example

User	Permission
Student	View Results
Lecturer	Enter Marks
Admin	Full Access

Only authorized users can access sensitive information.

3. Database Maintenance

Organizations use databases for many years.

Requirements may change over time.

The DBMS allows:

- Adding new tables
- Modifying existing structures
- Expanding database capacity

This process is called **Database Evolution**.

General-Purpose DBMS vs Customized Software

General-Purpose DBMS

Ready-made software designed for many applications.

Examples

- MySQL
- Oracle
- SQL Server

Advantages

- Easy to use
- Reliable
- Secure
- Supports multiple applications

Customized Database Software

Special software developed for a specific organization.

Example

- Airline Reservation System
- Hospital Management System
- Banking System

These systems may include their own specialized database programs.

1.2 An Example: UNIVERSITY Database

Introduction

To understand how a database works, consider a **UNIVERSITY Database**.

This database stores information about:

- Students
- Courses
- Course Sections
- Grades
- Prerequisites

The database is organized into **five files (tables)**.

```
CREATE DATABASE UNIVERSITY;  
  
USE UNIVERSITY;
```

2. Database Files (Tables)

2.1 STUDENT Table

Stores information about students.

Attribute	Description
Name	Student name
Student_number	Student ID
Class	Academic level
Major	Major subject

INSERT INTO STUDENT VALUES

```
(17,'Smith',1,'CS'),  
(8,'Brown',2,'CS');
```

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

```
CREATE TABLE STUDENT (  
    Student_number INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Class INT,  
    Major VARCHAR(20)  
);
```

Class Coding

Code	Meaning
1	Freshman
2	Sophomore
3	Junior
4	Senior
5	Graduate

2.2 COURSE Table

Stores information about courses.

Attribute	Description
Course_name	Course title
Course_number	Course code
Credit_hours	Credit value
Department	Offering department

CREATE TABLE COURSE (

Course_number VARCHAR(10) PRIMARY KEY,

Course_name VARCHAR(100),

Credit_hours INT,

Department VARCHAR(20)

);

INSERT INTO COURSE

VALUES

('CS1310','Intro to Computer Science',4,'CS'),
('CS3320','Data Structures',4,'CS'),
('MATH2410','Discrete Mathematics',3,'MATH'),
('CS3380','Database',3,'CS');

Course_number	Course_name	Credit_hours	Department
CS1310	Intro to Computer Science	4	CS
CS3320	Data Structures	4	CS
MATH2410	Discrete Mathematics	3	MATH
CS3380	Database	3	CS

2.3 SECTION Table

Stores information about course sections.

Attribute	Description
Section_identifier	Section ID
Course_number	Course code
Semester	Semester offered
Year	Academic year
Instructor	Teacher name

```
CREATE TABLE SECTION (  
    Section_identifier INT PRIMARY KEY,  
    Course_number VARCHAR(10),  
    Semester VARCHAR(20),  
    Year INT,  
    Instructor VARCHAR(50),  
    FOREIGN KEY (Course_number) REFERENCES COURSE(Course_number)  
);
```

INSERT INTO SECTION

VALUES

```
(85,'MATH2410','Fall',2007,'King'),  
(92,'CS1310','Fall',2007,'Anderson'),  
(102,'CS3320','Spring',2008,'Knuth'),  
(112,'MATH2410','Fall',2008,'Chang'),  
(119,'CS1310','Fall',2008,'Anderson'),  
(135,'CS3380','Fall',2008,'Stone');
```

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	2007	King
92	CS1310	Fall	2007	Anderson
102	CS3320	Spring	2008	Knuth
112	MATH2410	Fall	2008	Chang
119	CS1310	Fall	2008	Anderson
135	CS3380	Fall	2008	Stone

2.4 GRADE_REPORT Table

Stores students' grades.

Attribute	Description
Student_number	Student ID
Section_identifier	Section ID
Grade	Obtained grade

```
CREATE TABLE GRADE_REPORT (  
    Student_number INT,  
    Section_identifier INT,  
    Grade CHAR(1),  
    PRIMARY KEY(Student_number, Section_identifier),  
    FOREIGN KEY (Student_number) REFERENCES STUDENT(Student_number),  
    FOREIGN KEY (Section_identifier) REFERENCES SECTION(Section_identifier)  
);
```

INSERT INTO GRADE_REPORT

(Student_number, Section_identifier, Grade)

VALUES

(17,112,'B'),

(17,119,'C'),

(8,85,'A'),

(8,92,'A'),

(8,102,'B'),

(8,135,'A');

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

2.5 PREREQUISITE Table

Stores prerequisite courses.

Attribute	Description
Course_number	Course
Prerequisite_number	Required course

```
CREATE TABLE PREREQUISITE (  
    Course_number VARCHAR(10),  
    Prerequisite_number VARCHAR(10),  
    PRIMARY KEY(Course_number, Prerequisite_number),  
    FOREIGN KEY (Course_number) REFERENCES COURSE(Course_number),  
    FOREIGN KEY (Prerequisite_number) REFERENCES COURSE(Course_number)  
);
```

INSERT INTO PREREQUISITE

VALUES

('CS3380','CS3320'),

('CS3380','MATH2410'),

('CS3320','CS1310');

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310